



PRATEEK SRIVASTAVA

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CGPA/%
2023 - 2027	B.Tech. (Electrical Engineering)	Harcourt Butler Technical University, Kanpur (U.P.)	7.825/10
2022	Class XII (CBSE)	Kendriya Vidyalaya No.3 Kanpur (U.P.)	93.2%
2020	Class X (CBSE)	Kendriya Vidyalaya No.3 Kanpur (U.P.)	89.6%

ABOUT

I am final-year Electrical Engineering undergraduate at HBTU Kanpur specializing in Power Systems and Control Systems. My work bridges classical control theory with data-driven methods — applying Python and ML to predictive modeling and fault diagnostics across power systems.

INTERNSHIPS

Summer Intern | Stealth and Camouflage Department (SCD), DMSRDE-DRDO, Kanpur *Jun 2026 – Jul 2026*

- Completed a one-month research internship on "Measurement of Electromagnetic Properties of Materials at Microwave Frequencies", under Dr. Himangshu B. Baskey, Scientist 'F'.
- Operated a Vector Network Analyzer (VNA) for S-parameter (S11, S21) measurements in X-band (8.2–12.4 GHz) waveguide and SOLT calibration.
- Applied the Nicolson-Ross-Weir (NRW) algorithm to extract complex permittivity and permeability, validating results against a Teflon (PTFE) reference standard.
- Studied Frequency Selective Surfaces (FSS) and ABCD-matrix impedance matching for multilayer Radar Absorbing Material (RAM) design; gained exposure to CST simulation tools.

SKILLS

Core EE: Load Flow & Fault Analysis, Linear Control Theory, State-Space Analysis, Electrical Machines (Transformers, Synchronous Motors, Induction Drives).

AI & Software: Python; Predictive Modeling, Reinforcement Learning; NumPy, Pandas, Scikit-learn, Matplotlib.

Simulations: MATLAB/Simulink.

PROJECTS

Cost & Material Optimization Engine | [GitHub](#)

- I have developed a machine learning model to predict equipment costs based on load demand and spatial requirements.
- It is integrated with a recommendation system to suggest cost-effective materials, balancing durability with budget constraints.
- It reduced manual estimation time and improved cost accuracy through automated data-driven forecasting.

Sahaay AI – AI Farming Assistant (Group Project at Hackathon) | [GitHub](#)

Node.js, Python, FastAPI, Neo4j, Twilio, LangChain, Inngest

- AI assistant for farmers via WhatsApp/voice calls, no app needed; Inngest event-driven pipeline: Twilio → weather APIs → Tavily scraper → vector DB + Neo4j knowledge graph for crop-disease-treatment reasoning.

POSITION OR RESPONSIBILITIES

Design and Safety Head at Aeroclub HBTU *(2024-26)*

- Organized RC plane modeling and water rocket workshops.

Technical Head at Autorob HBTU *(2024-25)*

- Led technical training sessions for students on Gesture Control and Obstacle Avoiding robots.

ACHIEVEMENTS

Enigma Hackathon – RIET 2026

- Cleared 2 Rounds | Inter-college Hackathon, Jaipur, India.

Brainwave 2.0 – DTU 2026

- Cleared Round 1 | Presented Sahaay AI | Inter-college Hackathon, Delhi, India.

HOBBIES

- Singing